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## title: Homepage

### About

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I am a fourth-year PhD student in the Department of Mathematics at Brown University. My advisor is Prof. Terence Tao.

I completed my undergraduate studies at the University of Science and Technology of China (USTC).

I am interested in analysis and partial differential equations. My current research focuses on the Vlasov-Hartree system.

**Contact:** wenrui\_huang@brown.edu

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### Publications & Preprints

1. Long-Time Dynamics and Modified Scattering for the Coulomb Vlasov-Hartree System, (with M. Xie), arXiv:2608.04402 <details class="paper-abstract"> <summary><span class="visually-hidden">abstract</span></summary> <p>We study the three-dimensional Coulomb Vlasov-Hartree system, a classical-quantum model for Bose-Fermi mixtures. For small, regular, and localized data, the paper proves global well-posedness and identifies the coupled long-time dynamics: logarithmic corrections appear in both the Vlasov characteristics and the Hartree phase.</p> </details>
2. Scattering map for the Vlasov-Poisson system with a harmonic repulsive potential, (with H. Kwon), arXiv:2602.21344 <details class="paper-abstract"> <summary><span class="visually-hidden">abstract</span></summary> <p>We study scattering for the Vlasov-Poisson system with a repulsive harmonic potential in dimensions at least two. The work proves modified scattering and constructs wave operators through a lens-transform viewpoint, with weaker assumptions on the initial data than in some earlier treatments.</p> </details>
3. Stability and instability of small BGK waves, (with D. Bian, E. Grenier and B. Pausader), arXiv:2601.10030 <details class="paper-abstract"> <summary><span class="visually-hidden">abstract</span></summary> <p>This paper analyzes the linear behavior of small Bernstein-Green-Kruskal waves. The stability or instability mechanism is tied to the sign of the derivative of the energy distribution at zero energy.</p> </details>

4. The Vlasov-Poisson system with a perfectly conducting wall: Convex domains, (with B. Pausader and M. Suzuki), to appear in **Comm. Math. Phys.**   

Toggle abstract

We investigate the Vlasov-Poisson system in convex domains with a perfectly conducting boundary. Under localization and geometric assumptions, the asymptotic particle velocities are constrained by the asymptotic domain, and the dynamics exhibit a modified scattering behavior.
5. Modified wave operators for the Hartree equation with repulsive Coulomb potential, arXiv:2411.02737   

Toggle abstract

This work treats a final-state problem for the Hartree equation with repulsive Coulomb interaction. For small and sufficiently localized asymptotic data, it constructs a unique global solution with the prescribed modified scattering profile.
6. Scattering of the Vlasov-Riesz system in three dimensions, (with H. Kwon), **Annales Henri Poincaré** (2026), arXiv:2407.16919.   

Toggle abstract

We study long-time behavior for the three-dimensional Vlasov-Riesz system. The paper establishes scattering for small data in sub-Coulomb regimes, modified scattering near the Coulomb case, and corresponding wave operators, including polynomial corrections in the asymptotic description.

## Teaching

### Brown University

- Fall 2025 â TA â MATH 0190 Calculus II (Physics/Engineering)
- Fall 2023, Spring 2024, Fall 2026 â TA â MATH 0180 Calculus III
- Spring 2023 â TA â MATH 0100 Calculus II

### University of Science and Technology of China

- Spring 2022 â TA â Riemannian Geometry
- Spring 2021, Spring 2022 â TA â Topology
- Fall 2021 â TA â Differential Geometry

## Talks

- GLESPA seminar, Brown University, April 2026.
- Poster session in Workshop on Kinetic Theory and Fluids, University of Wisconsin-Madison, March 2025.
- GLESPA seminar, Brown University, April 2024.
- Brown-Yale PDE seminar, April 2024.

- Brown-Yale PDE seminar, November 2023.